AFC–7798

Sub. Code
1MBO4C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Botany

METHODS IN BIOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly the following terms.

1. AFLP
2. Gene expression
3. RAPD
4. Units of radio activity
5. Photo voltaic cells
6. Mass spectroscopy
7. Attribute
8. Biological database
9. Magnification
10. Condenser

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is RFLP? Describe its principle.

Or

(b) What are the advantages of DNA micro array technology in genomic analysis?

12. (a) Describe the functions of Geiger-Muller counter.

Or

(b) Explain RIA.

13. (a) Discuss the uses and applications of spectrophotometer.

Or

(b) Discuss the technique of mass spectroscopy.

14. (a) What is EMBL nucleotide sequence database? Mention the various link facilities available with EMBL.

Or

(b) Explain the concept of sequence alignment with its significance.

15. (a) Explain the resolving powers of different microscope.

Or

(b) Briefly explain about TEM.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Define electrophoresis. What are the procedure followed in the electrophoretic separation of DNA and proteins?

17. Describe the technique of ELISA.

18. Explain the methodology and role of X-ray diffraction.

19. Describe the different classes of Biological databases with its importance.

20. Describe the principle of electron microscopy and add a note on transmission electron microscope.

AFC–7782

Sub. Code
1MBO1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY – II
(PTERIDOPHYTES, GYMNOSPERMS AND PALEOBOTANY)

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Quilworts.
2. Vascular cryptogams.
3. Eustele.
4. Homosporous fern.
5. Paleozoic era.
6. Birbal Sahni.
7. *Anabaena cycadacearum*.
8. Bars of Sanio.
9. Geological Time scale.
10. Carbon dating.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the general characters of Rhyniales.

Or

(b) Explain the organization of spore producing organs of Calamitales.

12. (a) Give an account of seed habit in Pteridophytes.

Or

(b) Discuss the spore germination patterns as observed in Pteridophytes.

13. (a) Trace the origin of gymnosperms.

Or

(b) Explain the external structure of Pteridospermales.

14. (a) Discuss the organization of male strobili of Cycadales.

Or

(b) Examine the internal structure of stem of Coniferales.

15. (a) Analyse the internal structure of the stem of *Sphenophyllum*.

Or

(b) Give an account of paleopalynology.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the reproduction of Lycopodiales.

17. Write an essay on the soral evolution of pteridophytes.

18. Compare the external structure of the members of Pentoxylales.

19. Discuss the organization of cones in Gnetales.

20. Write an essay on fossilization.

AFC–7784

Sub. Code
1MBO1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

GENETICS, PLANT BREEDING AND BIOSTATISTICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. First filial generation.

2. Test cross.

3. Transversion.

4. Mutants.

5. Pseudoalleles.

6. Transposition.

7. Pure line.

8. High yielding variety.

9. Standard error.

10. Degree of freedom.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a brief account on nutritional and environmental theory of sex determination.

Or

(b) Write notes on the inheritance of gene for color blindness in man.

12. (a) Write a key note on physical mutagens.

Or

(b) Analyse the mutants of Neurospora.

13. (a) Write notes on plastid inheritance.

Or

(b) Write brief account on Kappa particles of paramecium.

14. (a) What are the advantages and limitations of plant breeding?

Or

(b) Write an account on the role of polyploidy in crop improvement.

15. (a) What do you mean by Binomial distribution of probability?

Or

(b) Analyse the features of normal distribution curve.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. The first white eyed mutant discovered in Drosophila was a male fly. Was this expected, or would you expect a fly of opposite sex? Why?

17. Write notes on

    (a) Chemical mutagen

    (b) Ionizing radiation

    (c) Lethal mutation

    (d) Amber mutant

    (e) Somatic mutation.

18. Mention Hardy Weinberg theory. Explain how different factors affect gene frequency in a population.

19. Discuss the various statistical test for determining the test of significance.

20. What are the objectives of sampling? Discuss the different types of sampling.

AFC–7785

Sub. Code
1MBO1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

Elective — MUSHROOM CULTIVATION AND VERMICULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Comment/Explain/Short note on the following :

1. Ascus.
2. Parasites.
3. Edible mushroom.
4. Spawn.
5. Mushroom pickles.
6. Worm cast.
7. Excretion of earthworm.
8. Nephridium.
9. Spermatophores.
10. Vermiwash.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Critically comment on poisonous mushroom.

Or

(b) Discuss the importance of mushroom.

12. (a) Give an account on the economics of mushroom cultivation.

Or

(b) Discuss the management of bacterial disease in mushroom cultivation.

13. (a) What are the different types of earthworm?

Or

(b) Elucidate briefly the north Indian species of earthworm.

14. (a) Narrate the pot method of vermicomposting.

Or

(b) What are the requirements for the preparation of vermicomposting?

15. (a) Discuss the role of earthworm in solid waste management.

Or

(b) Mention the uses of earthworm in land improvement and reclamation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the method of cultivation of white button mushroom.

17. Illustrate the principle and the technique of composting.

18. Explain the morphological and anatomical characteristics of earthworms.

19. Discuss the nutrient value of vermicomposting.

20. Give an account on the earthworm as bioreactors.

AFC–7786

Sub. Code
1MBO1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany – Elective

ORGANIC AGRICULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Hedge plant.
2. Green pasture.
3. Soil biota.
4. Bioferlilizers.
5. Multiple cropping.
6. Integrated farming.
7. Bioweedicides
8. Pheromone traps.
9. Synseeds.
10. IPR.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the future prospects of organic farming?

Or

(b) Comment on the need for land and water management.

12. (a) Outline the factors that effect soil fertility.

Or

(b) How can we accelerate the process of nutrient recycling?

13. (a) Comment on the choice of intercrops in relation to maintenance of soil fertility.

Or

(b) Outline the advantages and relay cropping system.

14. (a) What are the biological means of weed control?

Or

(b) Discuss the role of biopesticides in disease management.

15. (a) Comment on the market potential of organic produce.

Or

(b) Outline the procedure involved in the accreditation of organic produce.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Outline the principles and practices of organic farming for sustainable crop production.

17. Discuss the role of microbes in degradation and decomposition of organic residues.

18. Explain the role of intercropping in regulation/ management of soil nutrient status.

19. Explain the concept of integrated pest management.

20. What is the current senario of organic produce in the international market?

AFC–7787

Sub. Code
1MBO2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

TAXONOMY OF ANGIOSPERMS AND ETHANOBOTANY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 =20)

Answer **all** questions.

1. Name the families of *Gymnospermae.*

2. Comment on Phylogenetic system of classification.

3. Comment on Biosystematics.

4. Give a note on cluster analysis.

5. Differentiate rejection and retension.

6. What do you understand by type specimen?

7. Trace the floral features of *Annonaceae.*

8. Bring about the habit of *Aizoaceae.*

9. Write the salient features of *Scrophulariaceae.*

10. Mention any three distinquistny features of *Rubiaceae.*

Part B $(5 \times 5 = 25)$

Answer **all** questions, by choosing either (a) or (b).

11. (a) Write a critical note on the main concepts in Engler and Prantl's system of classification of Angiosperms.

Or

(b) What is the relationship between phylogency and classification of flowering plants.

12. (a) What are the contributions and functions of national and international herbaria to modern Toxonomical studies?

Or

(b) Outline the attributes of monographs in the field of plant systematics.

13. (a) Why International rules of nomenclature are important and when were they pronounced? Name the main principles governing the rules.

Or

(b) Indicate the role of Embryology in relation to taxonomy.

14. (a) Describe the range of floral structure in *Rosaceae*.

Or

(b) With the help of suitable illustration and formula, outline the floral characters of *Meliaceae*.

15. (a) Mention the distinquishing features of *Rubiaceae*. Write a key note on its fruits and seeds.

Or

(b) Write the systematic position of *Orchidaeae*. Write the salient features of the same.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the role of chemotaxonomy/phytochemistry in the classification and the determination of interrelationships.

17. Discuss the Taxonomic literature.

18. Enumerate the history of Botanical Nomenclature.

19. Compare the detailed floral features of *Capparidaceae* with that of *Sapindaceae*.

20. Describe the floral and diagnostic characters of *Poaceae*. List out the economic importance of *poaceae*.

AFC–7788

Sub. Code
1MBO2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

ANATOMY, EMBRYOLOGY AND MORPHOGENESIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Quiescent centre.

2. Vesselless angiosperms.

3. Periderm.

4. Leaf abscission.

5. Sporopollenin.

6. Triple fusion.

7. Helobial type.

8. Mosaic endosperm.

9. Morphogenesis.

10. Actin.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Distinguish between vascular cambium and cork cambium.

Or

(b) Distinguish between sieve tube and sieve cells.

12. (a) Write briefly on Tunica-Corpus concept.

Or

(b) Show how leaf traces are different from leaf gaps.

13. (a) Enumerate the methods to overcome incompatibility.

Or

(b) Write notes on the nutrition of embryosac.

14. (a) Write the significance of polyembryony.

Or

(b) List the difference between monocot and dicot embryo.

15. (a) Comment on the morphogenetic factors.

Or

(b) Critically comment on differentiation, dedifferentiation and redifferentiation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If xylem which consists of tubular extensions, of xylem vessels can conduct water upwards, why not the phloem which has sieve tubes do the same functions.

17. Explain the secondary structure of dicot stem with suitable example.

18. Write the organisation of mature embryosac.

19. Describe the development of monocot embryo.

20. Give an account on cytoskeleton.

AFC–7789

Sub. Code
1MBO2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Second Semester

Botany

MOLECULAR BIOLOGY AND EVOLUTION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define / Explain briefly the following terms :

1. Wobble.
2. Splicing.
3. Enhancer.
4. Distal Promoter.
5. Heat Shock Proteins.
6. Mutants.
7. Evolution.
8. Phylogenetic tree.
9. Sympatric speciation.
10. Genetic drift.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write notes on topoisomerases.

Or

(b) Describe the properties of genetic code.

12. (a) Explain the basic plant gene structure.

Or

(b) Write notes on DNA packaging.

13. (a) Write notes on recombinational repair.

Or

(b) Discuss the mechanism of chemically induced mutation.

14. (a) Describe gene divergence.

Or

(b) Write notes on protein sequence analysis.

15. (a) Write notes on allopatni speciation.

Or

(b) Write notes on random genetic drift.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the replication of DNA.

17. Explain the interaction of promoter sequence elements and transcription factors in plant gene expression.

18. Draw and describe the Holliday model of recombination.

19. Write an essay on isolating mechanisms.

20. Write an essay on process of speciation.

AFC–7790

Sub. Code
1MBO2C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

CELL COMMUNICATION AND CELL SIGNALING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Transformation.
2. Colorization.
3. G-proteins.
4. Quorum sensing.
5. Hematopoiesis.
6. Gap junctions.
7. Biological clock.
8. Protein sorting.
9. Phytochormes.
10. Phytotoxins.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the process of entry of bacteria into host cells.

Or

(b) Write a note on virus induced cell transformation.

12. (a) Comment on the role of secondary messengers in signal transduction.

Or

(b) Differentiate chemotaxsis and quorum sensing.

13. (a) Elucidate the principles of cell communication.

Or

(b) Explain the role of acetyl choline in neurotransmission.

14. (a) Discuss the role of phytochromes in the regulation of stomatal movements in plants.

Or

(b) Discuss the role of fungal elicitors in synthesis of secondary metabolites.

15. (a) Describe the techniques involved in the detection and quantization of phytohormons.

Or

(b) Comment on the role of ABA on plant growth and development in relation to the process of aging.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about pathogen induced diseases in plants.

17. With suitable diagram explain the two component system in plants.

18. Describe the role of voltage gated ion channels in neurotransmission.

19. Explain the process of protein trafficking from ER to GB.

20. Explain the physiological effects and modes operendii of phytohormones.

AFC–7791

Sub. Code
1MBO2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

Elective — ALGAL TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Microalgae.
2. Sterilization.
3. Photobioreactor.
4. Synthetic media.
5. Edible seaweeds.
6. Brown seaweeds.
7. Line method.
8. Vegetative method.
9. Agarophytes.
10. Nori.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Outline the general principles of culturing algae in laboratory.

Or

(b) Give the chemical composition of culture media for a fresh water algae.

12. (a) What the advantages of photo bioreactor over the conventional bioreactor in cultivating microalgae?

Or

(b) With diagrammatic sketch describe different types of photobioreactors.

13. (a) Elucidate the importance of light as a factor for the cultivation of microalgae.

Or

(b) Enlist the polysaccharides obtained from the seaweeds add a note on its application.

14. (a) Enlist the factors that affects the site selection in seaweed farming.

Or

(b) Prospect the market potential of seaweed produce.

15. (a) Bring out the importance of seaweed as a source of human food.

Or

(b) Describe the commercial application of seaweed polysaccharides.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. With systematics, describe any two fresh water microalgae.

17. Comment on the harvesting method with reference to oil extraction.

18. Bring out the importance of sexuality and sex attraction in seaweeds.

19. Write in detail about the harvesting process used to harvest the seaweeds.

20. Prospect the role of microalgae as a source of biofertilizer.

AFC–7792

Sub. Code
1MBO3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

ECOLOGY, BIOGEOGRAPHY AND CONSERVATION BIOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Climax community.
2. Allelopathy.
3. Thermocline.
4. Ecospecies.
5. Desertification.
6. Ozone depletor.
7. Red Data Book.
8. Continental Drift.
9. Monoculture operation.
10. Sacred groves.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Enumerate the characteristic features of population.

Or

(b) Write notes on the methods of studying plant community.

12. (a) How does different types of pollution affect the biochemical cycles of the environment?

Or

(b) Comment on the ecological interests of man made ecosystem with suitable example.

13. (a) What is thermal pollution and how can it be prevented?

Or

(b) What are the potential environmental impact of elevated atmospheric CO_2?

14. (a) Enumerate the biogeographical classification of India.

Or

(b) Write short notes on remote sensing and its significance in ecological studies.

15. (a) Write notes on the practice of silviculture.

Or

(b) Comment on the possible level of environmental education in the country.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the analytic and synthetic characteristics utilized for the study of plant community.

17. What is an ecosystem? Explain its diversity and stability.

18. Should India go for constructing mega dams? Discuss.

19. Discuss the continuous and discontinuous distribution of vegetation.

20. Discuss the steps of Indian government to conserve endangered and endemic species.

AFC–7793

Sub. Code
1MBO3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT PHYSIOLOGY AND BIOPHYSICAL CHEMISTRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short note only :

1. Diffusion Pressure Deficit (DPD).
2. Hydroponics.
3. Resonance transfer.
4. Krant'z anatomy.
5. Leghaemoglobin.
6. Nitrate Reductase (NR).
7. Bioassay.
8. Florigen.
9. Vander Waals Force.
10. Assimilatory power.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Comment on plant cell as Osmotic system.

Or

(b) What are trace elements? Explain their role in plant nutrition.

12. (a) What is dark reaction in photosynthesis? How does it take place in C_3 plants?

Or

(b) Point out the difference between photophosporylation and oxidative phosphorylation.

13. (a) On chopping over and again the hedge plants dense. Why? Explain.

Or

(b) What are phytoregulators? Give an account of the physiological role of cytokinin.

14. (a) How are nitrites converted to ammonia?

Or

(b) Write notes on nitrogen fixing organisms.

15. (a) How hydrogen bond is formed? Add a note on its biological significance.

Or

(b) Write notes on pH and its significance.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. What is water stress? How do plants respond to water stress?

17. Explain how Krant's anatomy facilitate the two corboxylation reactions occur in the chloroplast of mesophyll and bundle sheath cells.

18. Explain the role of nitrogen and the nitrogen metabolism in plants.

19. What are the conditions required for vernalization? Explain the mechanism of vernalization.

20. Compare the mitochondrial bioenergetics with chloroplast bioenergetics.

AFC–7794

Sub. Code
1MBO3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT BIO-TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Alkaline phosphatase.
2. Transposons.
3. Disarmed plasmid.
4. Reporter gene.
5. Wound Callus tissue.
6. Fusogen agent.
7. Amylase.
8. Gobar gas.
9. Biosafety.
10. Copy right.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the scope and future potential of biotechnology.

Or

(b) Explain the role of reverse transcriptase in gene cloning.

12. (a) Briefly describe the organization of Ti plasmid.

Or

(b) Give an account of micro projectile bombardment method of gene transfer.

13. (a) Describe the protocol and discuss the significance of callus culture.

Or

(b) Describe the protocol for isolation of protoplast from leaf.

14. (a) Describe the production of citric acid.

Or

(b) Write short notes on biohydrogen.

15. (a) What are the opposing views on the introduction on GM crops?

Or

(b) Elaborate trade secret.

Part C

(3 × 10 = 30)

Answer any **three** of the following not exceeding 4 pages each.

16. Describe the principle, procedure, limitations and applications of PCR techniques.

17. Explain the steps involved in the regeneration of transgenic plants through *Agrobacterium* mediated gene transfer.

18. Describe the method for the development of haploids from anther culture.

19. Describe the various substances and the different digester used for biogas production.

20. Describe the various forms of protection of Intellectual properties and discuss the benefits and disadvantages from IPR regime.

AFC–7795

Sub. Code
1MBO3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany – Elective

BIORESOURCE TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly, the following terms.

1. Natural Resource
2. Agro forestry
3. Solar Cells
4. Functions of wind turbine
5. Gas hydrates
6. Moderator in a nuclear reactor
7. Biogas
8. Sources of Biodisel
9. Biofuel cells
10. Combustion

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the benefits of Agro forestry.

Or

(b) Write notes on components of the Atmosphere.

12. (a) Describe the solar cell panel.

Or

(b) Explain the advantages of wind energy.

13. (a) Write notes on ranking of coal.

Or

(b) Describe the composition of Petroleum.

14. (a) What is a biodisel? Describe the production of Biodisel.

Or

(b) Explain Biomass gasification method.

15. (a) What are the uses of Biogas?

Or

(b) Explain the Methane Fermentation Process.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain energy conservation and energy efficiency.

17. Describe tidal wave as a source of energy.

18. Write notes on Geothermal Energy and discuss the advantage of it.

19. Describe the classification of Biomass.

20. Explain the Biochemical method of energy production from Biomass.

AFC–7796

Sub. Code
1MBO3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective – PLANT TISSUE CULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Sodium hypochloride
2. Dry heat sterilization
3. Spinning culture
4. Caulorhizogenesis
5. Clonal propagation
6. Organoids
7. Colchicine
8. Plasmolyticum
9. Elicitors
10. Bitransformation

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Give an outline of a plant tissue culture laboratory.

Or

(b) What is aseptic culture technique?

12. (a) Elaborate the role of hormones on cyto differentiation.

Or

(b) Write brief note on crown gall disease.

13. (a) Write an account on organogenesis and its applications.

Or

(b) Describe the protocol for preparing artificial seeds.

14. (a) What is ovary culture? Give an outline of the principle and ovary culture technique.

Or

(b) Describe the procedure of chemofusion.

15. (a) How are secondary metabolites produced through *in vitro*?

Or

(b) Briefly describe the methods of cryopreservation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the process of media preparation.

17. Give an outline of the methods of single cell culture.

18. Define somaclonal variation? Discuss the possible mechanisms causing somaclonal variation.

19. Describe the method for the development of androgenic haploids from anther and pollen culture.

20. Discuss in detail the applications of micro propagation in horticulture and forestry.

AFC–7797

Sub. Code
1MBO4C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Botany

RESEARCH METHODOLOGY

(CBCS – 2011 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly the following terms.

1. Research.
2. Review of Literature.
3. Reprints.
4. Monograph.
5. Tabulation.
6. Mean.
7. Data.
8. Citation.
9. Microprocessor.
10. www.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Enumerate the criteria for the selection of a problem.

Or

(b) Discuss in brief the layout of a research report.

12. (a) Discuss briefly the overview of a Review paper.

Or

(b) Discuss briefly the format of thesis preparation.

13. (a) Explain the advantages of graphic representation.

Or

(b) Describe the measure of central tendency.

14. (a) Discuss the methods of preparation of Histogram.

Or

(b) Describe various web browsers.

15. (a) List the characteristics features of a computer.

Or

(b) Describe the key features of internal memory.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Enumerate the sources of Review of Literature and suggest the use of library for this purpose.

17. Describe the types of research publication.

18. Explain the classification of Data.

19. Write note on (a) full paper (b) short communication

20. Explain the features of external memory and comment on information retrieval on net.

AFC–7781

Sub. Code
1MBO1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY — I

[ALGAE, FUNGI, LICHENS AND BRYOPHYTES]

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Describe/Write short notes/Define the following.

1. Auxospore.
2. Spirulina platensis.
3. Fritschiella.
4. Oxidation pond.
5. Oidia.
6. Perithecium.
7. Oogoniol.
8. Corticolous.
9. Peristome.
10. KASHYAP.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Give the salient features of Red algae.

Or

(b) Write a key note on symbiotic algae.

12. (a) Outline the phylogeny and interrelationship of algae.

Or

(b) Write a concise account on pollution indicating algae.

13. (a) Write notes on the diagnostic features of Zygomycotina.

Or

(b) Enumerate the general characters of class Deuteromycetes.

14. (a) Write an account on the somatic organization of Lichen.

Or

(b) Write a key note on hormonal control of sex organ development in fungi.

15. (a) List out the general characters of Marchantiales.

Or

(b) Write the interesting features of the spore dispersal mechanism in bryophytes.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the trend and criteria of algal classification proposed by Fritsch.

17. Write in detail on the culture procedure and cultivation methods of algae.

18. Write an essay on the economic importance of fungi.

19. Describe the phenomenon parasexuality in fungi.

20. Discuss the evolution of gametophyte and sporophyte in Bryophytes.

AFC–7782

Sub. Code
1MBO1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY – II
(PTERIDOPHYTES, GYMNOSPERMS AND PALEOBOTANY)

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Quilworts.
2. Vascular cryptogams.
3. Eustele.
4. Homosporous fern.
5. Paleozoic era.
6. Birbal Sahni.
7. *Anabaena cycadacearum*.
8. Bars of Sanio.
9. Geological Time scale.
10. Carbon dating.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the general characters of Rhyniales.

Or

(b) Explain the organization of spore producing organs of Calamitales.

12. (a) Give an account of seed habit in Pteridophytes.

Or

(b) Discuss the spore germination patterns as observed in Pteridophytes.

13. (a) Trace the origin of gymnosperms.

Or

(b) Explain the external structure of Pteridospermales.

14. (a) Discuss the organization of male strobili of Cycadales.

Or

(b) Examine the internal structure of stem of Coniferales.

15. (a) Analyse the internal structure of the stem of *Sphenophyllum*.

Or

(b) Give an account of paleopalynology.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the reproduction of Lycopodiales.

17. Write an essay on the soral evolution of pteridophytes.

18. Compare the external structure of the members of Pentoxylales.

19. Discuss the organization of cones in Gnetales.

20. Write an essay on fossilization.

AFC–7784

Sub. Code
1MBO1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

GENETICS, PLANT BREEDING AND BIOSTATISTICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. First filial generation.

2. Test cross.

3. Transversion.

4. Mutants.

5. Pseudoalleles.

6. Transposition.

7. Pure line.

8. High yielding variety.

9. Standard error.

10. Degree of freedom.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write a brief account on nutritional and environmental theory of sex determination.

Or

(b) Write notes on the inheritance of gene for color blindness in man.

12. (a) Write a key note on physical mutagens.

Or

(b) Analyse the mutants of Neurospora.

13. (a) Write notes on plastid inheritance.

Or

(b) Write brief account on Kappa particles of paramecium.

14. (a) What are the advantages and limitations of plant breeding?

Or

(b) Write an account on the role of polyploidy in crop improvement.

15. (a) What do you mean by Binomial distribution of probability?

Or

(b) Analyse the features of normal distribution curve.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. The first white eyed mutant discovered in Drosophila was a male fly. Was this expected, or would you expect a fly of opposite sex? Why?

17. Write notes on

    (a) Chemical mutagen

    (b) Ionizing radiation

    (c) Lethal mutation

    (d) Amber mutant

    (e) Somatic mutation.

18. Mention Hardy Weinberg theory. Explain how different factors affect gene frequency in a population.

19. Discuss the various statistical test for determining the test of significance.

20. What are the objectives of sampling? Discuss the different types of sampling.

AFC–7785

Sub. Code
1MBO1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

Elective — MUSHROOM CULTIVATION AND VERMICULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Comment/Explain/Short note on the following :

1. Ascus.
2. Parasites.
3. Edible mushroom.
4. Spawn.
5. Mushroom pickles.
6. Worm cast.
7. Excretion of earthworm.
8. Nephridium.
9. Spermatophores.
10. Vermiwash.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Critically comment on poisonous mushroom.

Or

(b) Discuss the importance of mushroom.

12. (a) Give an account on the economics of mushroom cultivation.

Or

(b) Discuss the management of bacterial disease in mushroom cultivation.

13. (a) What are the different types of earthworm?

Or

(b) Elucidate briefly the north Indian species of earthworm.

14. (a) Narrate the pot method of vermicomposting.

Or

(b) What are the requirements for the preparation of vermicomposting?

15. (a) Discuss the role of earthworm in solid waste management.

Or

(b) Mention the uses of earthworm in land improvement and reclamation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the method of cultivation of white button mushroom.

17. Illustrate the principle and the technique of composting.

18. Explain the morphological and anatomical characteristics of earthworms.

19. Discuss the nutrient value of vermicomposting.

20. Give an account on the earthworm as bioreactors.

AFC–7786

Sub. Code
1MBO1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany – Elective

ORGANIC AGRICULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Hedge plant.
2. Green pasture.
3. Soil biota.
4. Bioferlilizers.
5. Multiple cropping.
6. Integrated farming.
7. Bioweedicides
8. Pheromone traps.
9. Synseeds.
10. IPR.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the future prospects of organic farming?

Or

(b) Comment on the need for land and water management.

12. (a) Outline the factors that effect soil fertility.

Or

(b) How can we accelerate the process of nutrient recycling?

13. (a) Comment on the choice of intercrops in relation to maintenance of soil fertility.

Or

(b) Outline the advantages and relay cropping system.

14. (a) What are the biological means of weed control?

Or

(b) Discuss the role of biopesticides in disease management.

15. (a) Comment on the market potential of organic produce.

Or

(b) Outline the procedure involved in the accreditation of organic produce.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Outline the principles and practices of organic farming for sustainable crop production.

17. Discuss the role of microbes in degradation and decomposition of organic residues.

18. Explain the role of intercropping in regulation/ management of soil nutrient status.

19. Explain the concept of integrated pest management.

20. What is the current senario of organic produce in the international market?

AFC–7787

Sub. Code
1MBO2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

TAXONOMY OF ANGIOSPERMS AND ETHANOBOTANY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 =20)

Answer **all** questions.

1. Name the families of *Gymnospermae.*

2. Comment on Phylogenetic system of classification.

3. Comment on Biosystematics.

4. Give a note on cluster analysis.

5. Differentiate rejection and retension.

6. What do you understand by type specimen?

7. Trace the floral features of *Annonaceae.*

8. Bring about the habit of *Aizoaceae.*

9. Write the salient features of *Scrophulariaceae.*

10. Mention any three distinquistny features of *Rubiaceae.*

Part B $(5 \times 5 = 25)$

Answer **all** questions, by choosing either (a) or (b).

11. (a) Write a critical note on the main concepts in Engler and Prantl's system of classification of Angiosperms.

Or

(b) What is the relationship between phylogency and classification of flowering plants.

12. (a) What are the contributions and functions of national and international herbaria to modern Toxonomical studies?

Or

(b) Outline the attributes of monographs in the field of plant systematics.

13. (a) Why International rules of nomenclature are important and when were they pronounced? Name the main principles governing the rules.

Or

(b) Indicate the role of Embryology in relation to taxonomy.

14. (a) Describe the range of floral structure in *Rosaceae*.

Or

(b) With the help of suitable illustration and formula, outline the floral characters of *Meliaceae*.

15. (a) Mention the distinquishing features of *Rubiaceae*. Write a key note on its fruits and seeds.

Or

(b) Write the systematic position of *Orchidaeae*. Write the salient features of the same.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the role of chemotaxonomy/phytochemistry in the classification and the determination of interrelationships.

17. Discuss the Taxonomic literature.

18. Enumerate the history of Botanical Nomenclature.

19. Compare the detailed floral features of *Capparidaceae* with that of *Sapindaceae*.

20. Describe the floral and diagnostic characters of *Poaceae*. List out the economic importance of *poaceae*.

AFC–7788

Sub. Code
1MBO2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

ANATOMY, EMBRYOLOGY AND MORPHOGENESIS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Quiescent centre.

2. Vesselless angiosperms.

3. Periderm.

4. Leaf abscission.

5. Sporopollenin.

6. Triple fusion.

7. Helobial type.

8. Mosaic endosperm.

9. Morphogenesis.

10. Actin.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Distinguish between vascular cambium and cork cambium.

Or

(b) Distinguish between sieve tube and sieve cells.

12. (a) Write briefly on Tunica-Corpus concept.

Or

(b) Show how leaf traces are different from leaf gaps.

13. (a) Enumerate the methods to overcome incompatibility.

Or

(b) Write notes on the nutrition of embryosac.

14. (a) Write the significance of polyembryony.

Or

(b) List the difference between monocot and dicot embryo.

15. (a) Comment on the morphogenetic factors.

Or

(b) Critically comment on differentiation, dedifferentiation and redifferentiation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If xylem which consists of tubular extensions, of xylem vessels can conduct water upwards, why not the phloem which has sieve tubes do the same functions.

17. Explain the secondary structure of dicot stem with suitable example.

18. Write the organisation of mature embryosac.

19. Describe the development of monocot embryo.

20. Give an account on cytoskeleton.

AFC–7789

Sub. Code
1MBO2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015.

Second Semester

Botany

MOLECULAR BIOLOGY AND EVOLUTION

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define / Explain briefly the following terms :

1. Wobble.
2. Splicing.
3. Enhancer.
4. Distal Promoter.
5. Heat Shock Proteins.
6. Mutants.
7. Evolution.
8. Phylogenetic tree.
9. Sympatric speciation.
10. Genetic drift.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write notes on topoisomerases.

Or

(b) Describe the properties of genetic code.

12. (a) Explain the basic plant gene structure.

Or

(b) Write notes on DNA packaging.

13. (a) Write notes on recombinational repair.

Or

(b) Discuss the mechanism of chemically induced mutation.

14. (a) Describe gene divergence.

Or

(b) Write notes on protein sequence analysis.

15. (a) Write notes on allopatni speciation.

Or

(b) Write notes on random genetic drift.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the replication of DNA.

17. Explain the interaction of promoter sequence elements and transcription factors in plant gene expression.

18. Draw and describe the Holliday model of recombination.

19. Write an essay on isolating mechanisms.

20. Write an essay on process of speciation.

AFC–7790

Sub. Code
1MBO2C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

CELL COMMUNICATION AND CELL SIGNALING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Transformation.
2. Colorization.
3. G-proteins.
4. Quorum sensing.
5. Hematopoiesis.
6. Gap junctions.
7. Biological clock.
8. Protein sorting.
9. Phytochormes.
10. Phytotoxins.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the process of entry of bacteria into host cells.

Or

(b) Write a note on virus induced cell transformation.

12. (a) Comment on the role of secondary messengers in signal transduction.

Or

(b) Differentiate chemotaxsis and quorum sensing.

13. (a) Elucidate the principles of cell communication.

Or

(b) Explain the role of acetyl choline in neurotransmission.

14. (a) Discuss the role of phytochromes in the regulation of stomatal movements in plants.

Or

(b) Discuss the role of fungal elicitors in synthesis of secondary metabolites.

15. (a) Describe the techniques involved in the detection and quantization of phytohormons.

Or

(b) Comment on the role of ABA on plant growth and development in relation to the process of aging.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about pathogen induced diseases in plants.

17. With suitable diagram explain the two component system in plants.

18. Describe the role of voltage gated ion channels in neurotransmission.

19. Explain the process of protein trafficking from ER to GB.

20. Explain the physiological effects and modes operendii of phytohormones.

AFC–7791

Sub. Code
1MBO2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Botany

Elective — ALGAL TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Microalgae.
2. Sterilization.
3. Photobioreactor.
4. Synthetic media.
5. Edible seaweeds.
6. Brown seaweeds.
7. Line method.
8. Vegetative method.
9. Agarophytes.
10. Nori.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Outline the general principles of culturing algae in laboratory.

Or

(b) Give the chemical composition of culture media for a fresh water algae.

12. (a) What the advantages of photo bioreactor over the conventional bioreactor in cultivating microalgae?

Or

(b) With diagrammatic sketch describe different types of photobioreactors.

13. (a) Elucidate the importance of light as a factor for the cultivation of microalgae.

Or

(b) Enlist the polysaccharides obtained from the seaweeds add a note on its application.

14. (a) Enlist the factors that affects the site selection in seaweed farming.

Or

(b) Prospect the market potential of seaweed produce.

15. (a) Bring out the importance of seaweed as a source of human food.

Or

(b) Describe the commercial application of seaweed polysaccharides.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. With systematics, describe any two fresh water microalgae.

17. Comment on the harvesting method with reference to oil extraction.

18. Bring out the importance of sexuality and sex attraction in seaweeds.

19. Write in detail about the harvesting process used to harvest the seaweeds.

20. Prospect the role of microalgae as a source of biofertilizer.

AFC–7792

Sub. Code
1MBO3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

ECOLOGY, BIOGEOGRAPHY AND CONSERVATION BIOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Climax community.
2. Allelopathy.
3. Thermocline.
4. Ecospecies.
5. Desertification.
6. Ozone depletor.
7. Red Data Book.
8. Continental Drift.
9. Monoculture operation.
10. Sacred groves.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Enumerate the characteristic features of population.

Or

(b) Write notes on the methods of studying plant community.

12. (a) How does different types of pollution affect the biochemical cycles of the environment?

Or

(b) Comment on the ecological interests of man made ecosystem with suitable example.

13. (a) What is thermal pollution and how can it be prevented?

Or

(b) What are the potential environmental impact of elevated atmospheric CO_2?

14. (a) Enumerate the biogeographical classification of India.

Or

(b) Write short notes on remote sensing and its significance in ecological studies.

15. (a) Write notes on the practice of silviculture.

Or

(b) Comment on the possible level of environmental education in the country.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the analytic and synthetic characteristics utilized for the study of plant community.

17. What is an ecosystem? Explain its diversity and stability.

18. Should India go for constructing mega dams? Discuss.

19. Discuss the continuous and discontinuous distribution of vegetation.

20. Discuss the steps of Indian government to conserve endangered and endemic species.

AFC–7793

Sub. Code
1MBO3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT PHYSIOLOGY AND BIOPHYSICAL CHEMISTRY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Write short note only :

1. Diffusion Pressure Deficit (DPD).
2. Hydroponics.
3. Resonance transfer.
4. Krant'z anatomy.
5. Leghaemoglobin.
6. Nitrate Reductase (NR).
7. Bioassay.
8. Florigen.
9. Vander Waals Force.
10. Assimilatory power.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Comment on plant cell as Osmotic system.

Or

(b) What are trace elements? Explain their role in plant nutrition.

12. (a) What is dark reaction in photosynthesis? How does it take place in C_3 plants?

Or

(b) Point out the difference between photophosporylation and oxidative phosphorylation.

13. (a) On chopping over and again the hedge plants dense. Why? Explain.

Or

(b) What are phytoregulators? Give an account of the physiological role of cytokinin.

14. (a) How are nitrites converted to ammonia?

Or

(b) Write notes on nitrogen fixing organisms.

15. (a) How hydrogen bond is formed? Add a note on its biological significance.

Or

(b) Write notes on pH and its significance.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. What is water stress? How do plants respond to water stress?

17. Explain how Krant's anatomy facilitate the two corboxylation reactions occur in the chloroplast of mesophyll and bundle sheath cells.

18. Explain the role of nitrogen and the nitrogen metabolism in plants.

19. What are the conditions required for vernalization? Explain the mechanism of vernalization.

20. Compare the mitochondrial bioenergetics with chloroplast bioenergetics.

AFC–7794

Sub. Code
1MBO3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

PLANT BIO-TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Alkaline phosphatase.
2. Transposons.
3. Disarmed plasmid.
4. Reporter gene.
5. Wound Callus tissue.
6. Fusogen agent.
7. Amylase.
8. Gobar gas.
9. Biosafety.
10. Copy right.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the scope and future potential of biotechnology.

Or

(b) Explain the role of reverse transcriptase in gene cloning.

12. (a) Briefly describe the organization of Ti plasmid.

Or

(b) Give an account of micro projectile bombardment method of gene transfer.

13. (a) Describe the protocol and discuss the significance of callus culture.

Or

(b) Describe the protocol for isolation of protoplast from leaf.

14. (a) Describe the production of citric acid.

Or

(b) Write short notes on biohydrogen.

15. (a) What are the opposing views on the introduction on GM crops?

Or

(b) Elaborate trade secret.

Part C

(3 × 10 = 30)

Answer any **three** of the following not exceeding 4 pages each.

16. Describe the principle, procedure, limitations and applications of PCR techniques.

17. Explain the steps involved in the regeneration of transgenic plants through *Agrobacterium* mediated gene transfer.

18. Describe the method for the development of haploids from anther culture.

19. Describe the various substances and the different digester used for biogas production.

20. Describe the various forms of protection of Intellectual properties and discuss the benefits and disadvantages from IPR regime.

AFC–7795

Sub. Code
1MBO3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany – Elective

BIORESOURCE TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly, the following terms.

1. Natural Resource
2. Agro forestry
3. Solar Cells
4. Functions of wind turbine
5. Gas hydrates
6. Moderator in a nuclear reactor
7. Biogas
8. Sources of Biodisel
9. Biofuel cells
10. Combustion

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the benefits of Agro forestry.

Or

(b) Write notes on components of the Atmosphere.

12. (a) Describe the solar cell panel.

Or

(b) Explain the advantages of wind energy.

13. (a) Write notes on ranking of coal.

Or

(b) Describe the composition of Petroleum.

14. (a) What is a biodisel? Describe the production of Biodisel.

Or

(b) Explain Biomass gasification method.

15. (a) What are the uses of Biogas?

Or

(b) Explain the Methane Fermentation Process.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain energy conservation and energy efficiency.

17. Describe tidal wave as a source of energy.

18. Write notes on Geothermal Energy and discuss the advantage of it.

19. Describe the classification of Biomass.

20. Explain the Biochemical method of energy production from Biomass.

AFC–7796

Sub. Code
1MBO3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Botany

Elective – PLANT TISSUE CULTURE

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Sodium hypochloride
2. Dry heat sterilization
3. Spinning culture
4. Caulorhizogenesis
5. Clonal propagation
6. Organoids
7. Colchicine
8. Plasmolyticum
9. Elicitors
10. Bitransformation

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Give an outline of a plant tissue culture laboratory.

Or

(b) What is aseptic culture technique?

12. (a) Elaborate the role of hormones on cyto differentiation.

Or

(b) Write brief note on crown gall disease.

13. (a) Write an account on organogenesis and its applications.

Or

(b) Describe the protocol for preparing artificial seeds.

14. (a) What is ovary culture? Give an outline of the principle and ovary culture technique.

Or

(b) Describe the procedure of chemofusion.

15. (a) How are secondary metabolites produced through *in vitro*?

Or

(b) Briefly describe the methods of cryopreservation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the process of media preparation.

17. Give an outline of the methods of single cell culture.

18. Define somaclonal variation? Discuss the possible mechanisms causing somaclonal variation.

19. Describe the method for the development of androgenic haploids from anther and pollen culture.

20. Discuss in detail the applications of micro propagation in horticulture and forestry.

AFC–7797

Sub. Code
1MBO4C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Botany

RESEARCH METHODOLOGY

(CBCS – 2011 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly the following terms.

1. Research.
2. Review of Literature.
3. Reprints.
4. Monograph.
5. Tabulation.
6. Mean.
7. Data.
8. Citation.
9. Microprocessor.
10. www.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Enumerate the criteria for the selection of a problem.

Or

(b) Discuss in brief the layout of a research report.

12. (a) Discuss briefly the overview of a Review paper.

Or

(b) Discuss briefly the format of thesis preparation.

13. (a) Explain the advantages of graphic representation.

Or

(b) Describe the measure of central tendency.

14. (a) Discuss the methods of preparation of Histogram.

Or

(b) Describe various web browsers.

15. (a) List the characteristics features of a computer.

Or

(b) Describe the key features of internal memory.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Enumerate the sources of Review of Literature and suggest the use of library for this purpose.

17. Describe the types of research publication.

18. Explain the classification of Data.

19. Write note on (a) full paper (b) short communication

20. Explain the features of external memory and comment on information retrieval on net.

AFC–7798

Sub. Code
1MBO4C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Fourth Semester

Botany

METHODS IN BIOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Define/Explain briefly the following terms.

1. AFLP
2. Gene expression
3. RAPD
4. Units of radio activity
5. Photo voltaic cells
6. Mass spectroscopy
7. Attribute
8. Biological database
9. Magnification
10. Condenser

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is RFLP? Describe its principle.

Or

(b) What are the advantages of DNA micro array technology in genomic analysis?

12. (a) Describe the functions of Geiger-Muller counter.

Or

(b) Explain RIA.

13. (a) Discuss the uses and applications of spectrophotometer.

Or

(b) Discuss the technique of mass spectroscopy.

14. (a) What is EMBL nucleotide sequence database? Mention the various link facilities available with EMBL.

Or

(b) Explain the concept of sequence alignment with its significance.

15. (a) Explain the resolving powers of different microscope.

Or

(b) Briefly explain about TEM.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Define electrophoresis. What are the procedure followed in the electrophoretic separation of DNA and proteins?

17. Describe the technique of ELISA.

18. Explain the methodology and role of X-ray diffraction.

19. Describe the different classes of Biological databases with its importance.

20. Describe the principle of electron microscopy and add a note on transmission electron microscope.

AFC–7781

Sub. Code
1MBO1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Botany

PLANT DIVERSITY — I

[ALGAE, FUNGI, LICHENS AND BRYOPHYTES]

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

Describe/Write short notes/Define the following.

1. Auxospore.
2. Spirulina platensis.
3. Fritschiella.
4. Oxidation pond.
5. Oidia.
6. Perithecium.
7. Oogoniol.
8. Corticolous.
9. Peristome.
10. KASHYAP.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Give the salient features of Red algae.

Or

(b) Write a key note on symbiotic algae.

12. (a) Outline the phylogeny and interrelationship of algae.

Or

(b) Write a concise account on pollution indicating algae.

13. (a) Write notes on the diagnostic features of Zygomycotina.

Or

(b) Enumerate the general characters of class Deuteromycetes.

14. (a) Write an account on the somatic organization of Lichen.

Or

(b) Write a key note on hormonal control of sex organ development in fungi.

15. (a) List out the general characters of Marchantiales.

Or

(b) Write the interesting features of the spore dispersal mechanism in bryophytes.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the trend and criteria of algal classification proposed by Fritsch.

17. Write in detail on the culture procedure and cultivation methods of algae.

18. Write an essay on the economic importance of fungi.

19. Describe the phenomenon parasexuality in fungi.

20. Discuss the evolution of gametophyte and sporophyte in Bryophytes.